

INFLATION

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THE (ELUSIVE) ROAD TOWARDS 2% INFLATION

Annualized PCE inflation through the year in March 2026:
5.6% headline, 4.2% excluding energy and 4.4% Core.

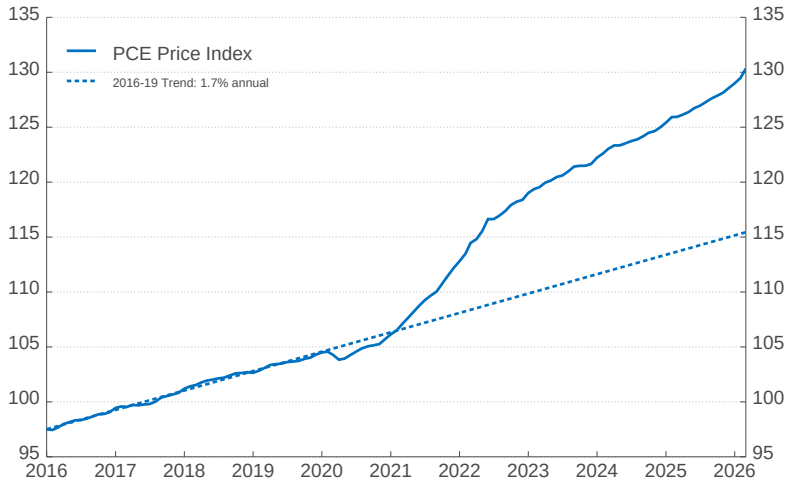
Inflation rates mostly unchanged for 2023, 2024 and 2025.

Notable increase in prices of core goods and core services ex. housing.

Prices and Inflation

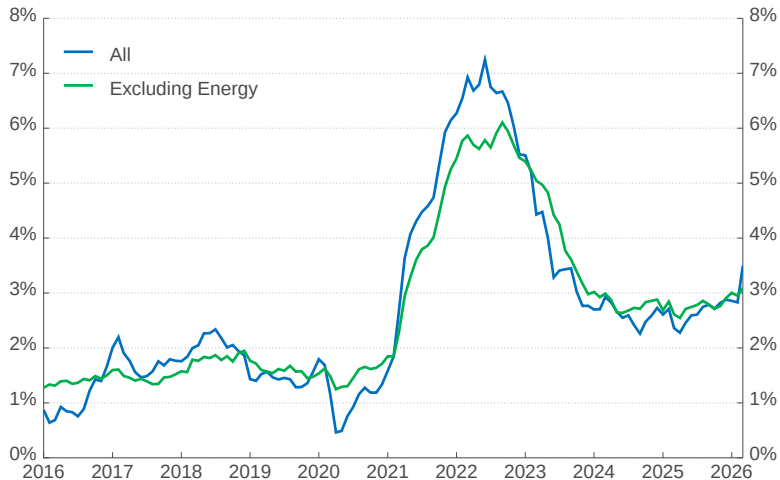
PRICES: STEADY INCREASE

INITIAL DECLINE DURING PANDEMIC—FAST INCREASE AFTERWARDS—ABOVE-TARGET GROWTH RATE SINCE 2023



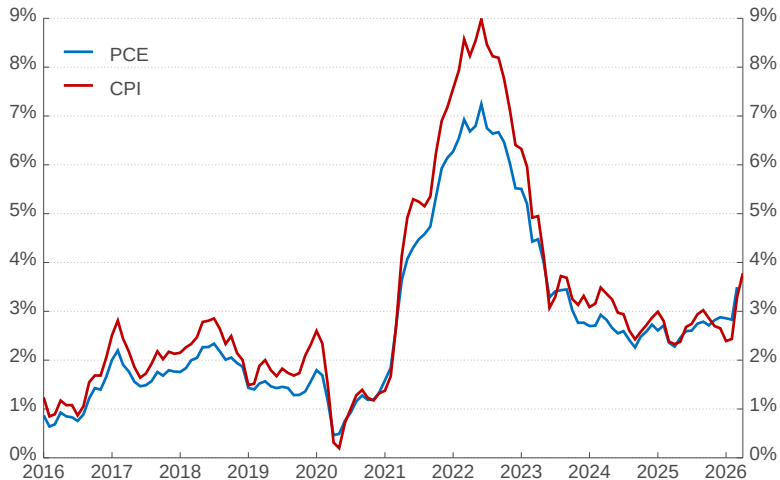
INFLATION: FAST DECLINE IN 2023, STALLING IN 2024-25

INFLATION HAS REMAINED ABOVE 2% SINCE MARCH 2021—DYNAMICS AMPLIFIED BY ENERGY ▷ HISTORICAL SERIES



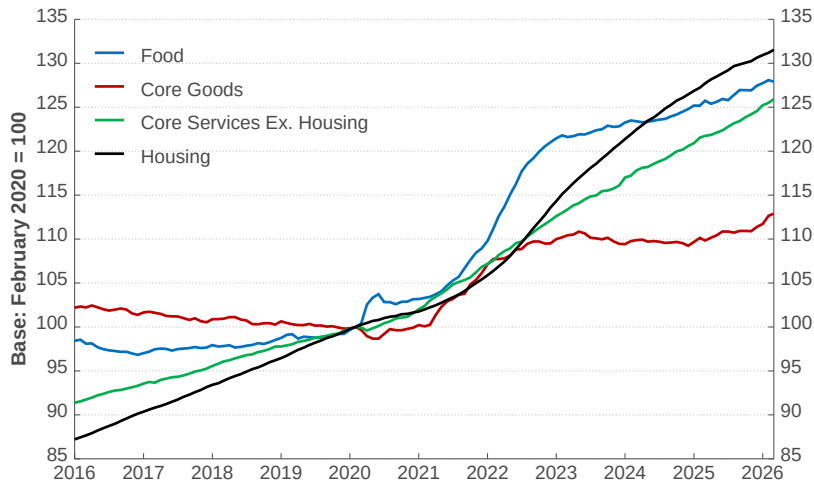
COMPARING DYNAMICS IN PCE AND CPI

LACK OF SUBSTANTIAL PROGRESS TOWARDS TARGET, WITH SOME RECENT DIVERGENCE ▷ EXCLUDING ENERGY



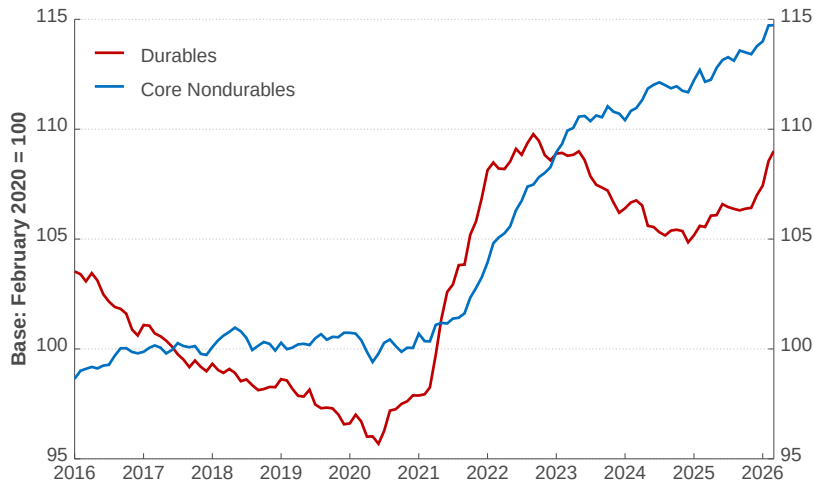
COMPONENTS OF PCE PRICE INDEX

EFFECT OF TARIFFS EVIDENT ON FOOD AND CORE GOODS PRICES



CORE GOODS: DURABLES VS. NONDURABLES

EFFECT OF TARIFFS EVIDENT ON PRICES OF CORE DURABLE GOODS—CORE NONDURABLES GROWING STEADILY



INFLATION REMAINS ABOVE THE FED'S TARGET

AGGREGATE INFLATION MEASURES—STRONG MODERATION IN 2023 BUT LACK OF PROGRESS IN 2024-25 ➤ CPI

Annualized PCE Inflation Rates

	All Goods & Services	Excluding Energy	Excluding Food & Energy (Core)
2016–19	1.7%	1.6%	1.7%
2020	1.3%	1.7%	1.5%
2021	6.1%	5.2%	5.2%
2022	5.5%	5.5%	5.0%
2023	2.8%	3.0%	3.1%
2024	2.7%	2.9%	3.0%
2025	2.9%	2.9%	3.0%
2026 YTD	5.6%	4.2%	4.4%

SOURCES: Bureau of Economic Analysis and author's calculations.

NOTE: The 2026 period is the year through March.

PANDEMIC DISRUPTED SOME PRE-EXISTING INFLATION TRENDS

STRONG REVERSALS IN FOOD, ENERGY AND CORE GOODS; MODERATION IN CORE SERVICES AND HOUSING ▸ CPI

Annualized PCE Inflation Rates by Categories

	Food	Energy	Core Goods	Core Services ex. Housing	Housing
2016–19	0.2%	4.2%	−0.6%	2.2%	3.4%
2020	4.0%	−7.5%	0.1%	2.0%	2.2%
2021	5.6%	30.3%	6.2%	5.2%	3.7%
2022	11.1%	6.7%	3.2%	4.9%	7.8%
2023	1.5%	−2.1%	0.0%	3.5%	6.4%
2024	1.6%	−1.0%	−0.2%	3.9%	4.7%
2025	2.1%	2.2%	2.0%	3.3%	3.3%
2026 YTD	1.4%	48.6%	5.5%	4.5%	2.8%

SOURCES: Bureau of Economic Analysis and author's calculations.

NOTE: The 2026 period is the year through March.

ESTIMATING IMPACT OF TARIFFS ON INFLATION

SIMPLE BACK-OF-THE-ENVELOPE CALCULATION VS. SOPHISTICATED MODELS.

Annual PCE inflation in 2025: 2.9%.

Compute inflation ex tariffs by using inflation rates from one year ago for food and core goods.

- ▶ Tariffs contributed 0.50 percentage points to inflation.
- ▶ Inflation excluding tariffs in 2025 estimated at 2.4%.
- ▶ Other indexes ex tariffs: Ex Energy at 2.4% and Core at 2.5%.

Lower estimates of impact when assuming core nondurables were not affected by tariffs: 0.40.

Thus, estimated inflation ex tariffs in 2025 was 2.4-2.5%.

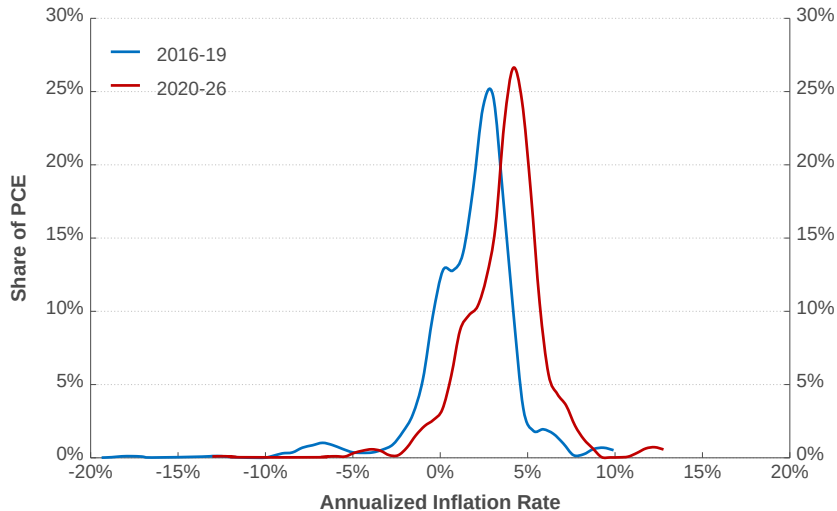
Model by STL FRB staff yields similar estimates of impact: 0.50 (headline) and 0.38 (Core).

Inflation Distributions

HIGH INFLATION EPISODE WAS WIDESPREAD

► METHODOLOGY

ESTIMATED DISTRIBUTION OF PCE INFLATION ► CORE



INFLATION DISTRIBUTIONS: PROGRESSION FROM 2016-19 TO 2026

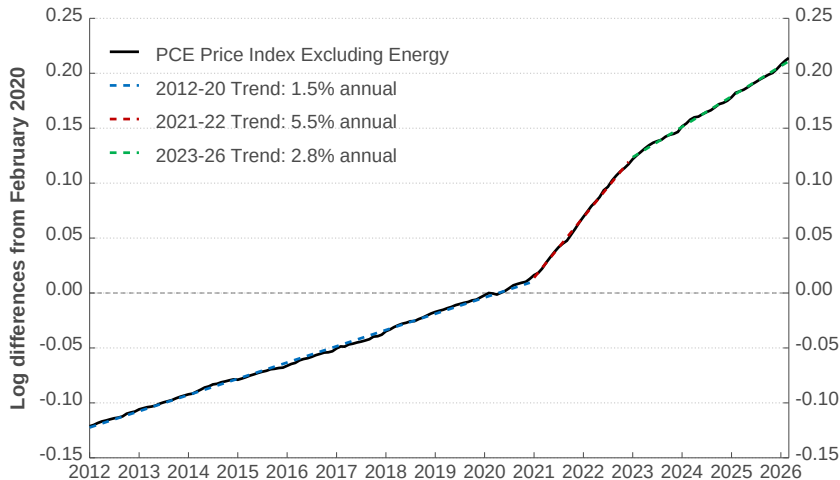
ESTIMATED DISTRIBUTION OF PCE INFLATION

Inflation Regimes

PRICE LEVEL DYNAMICS: THREE REGIMES

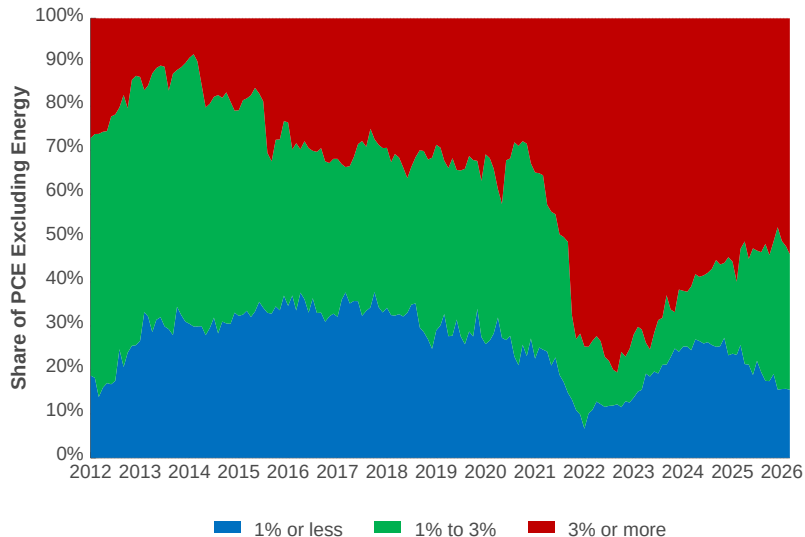
► [BLOG](#)

(1) BELOW-TARGET INFLATION: 2012-20 (2) GREAT INFLATION: 2021-22 (3) ABOVE-TARGET INFLATION: 2023-25



EXPENDITURE SHARES BY INFLATION RATES

ALMOST HALF OF CONSUMPTION EXPENDITURES ARE ON PRODUCTS WITH INFLATION RATES OF 3% OR MORE



LONG-TERM INFLATION EXPECTATIONS HAVE RISEN

MARKETS VS. MODELS VS. FORECASTERS

10-Year Inflation Expectations — Period Averages

	TIPS	Swaps	Cleveland	DKW	SPF
2012–20	1.9%	2.2%	1.7%	2.2%	2.0%
2021–22	2.4%	2.7%	1.9%	2.2%	2.3%
2023–26	2.3%	2.5%	2.3%	2.6%	2.2%
Change 2012-20 to 2023-26	0.4%	0.3%	0.5%	0.4%	0.1%

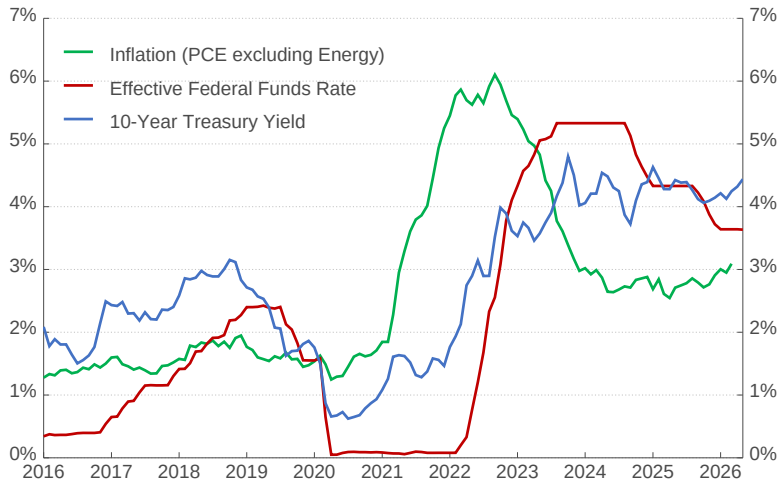
SOURCES: FRED, Federal Reserve Board, Federal Reserve Bank of Philadelphia, Bloomberg and author's calculations.

NOTE: The 2026 period is through April.

Interest Rates

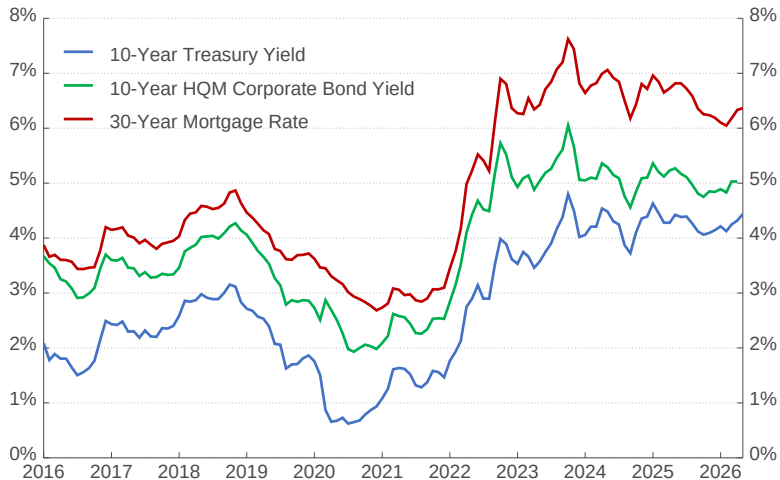
MONETARY POLICY

EASING CYCLES IN SEPTEMBER-DECEMBER 2024 AND SEPTEMBER-DECEMBER 2025



MONETARY POLICY: IMPACT ON LONG-TERM RATES

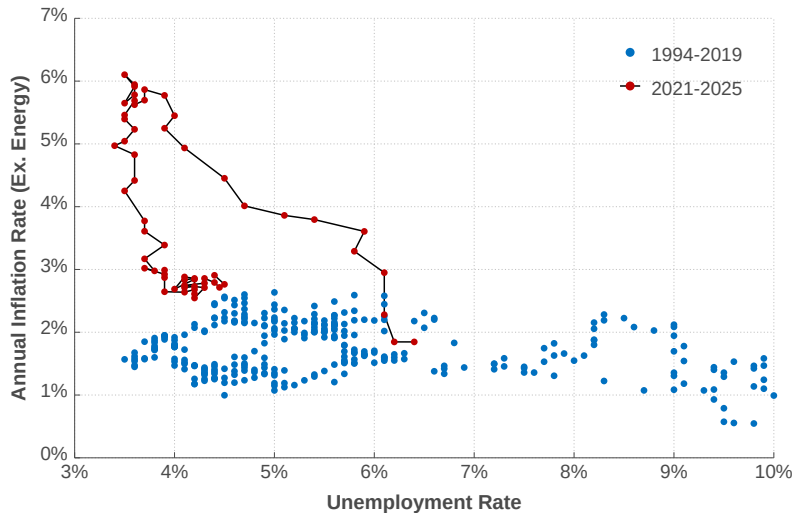
LONG-TERM RATES HAVE NOT YET RESPONDED TO EASING CYCLE—BUT RECENT DECLINE IN MORTGAGES



Phillips Curve

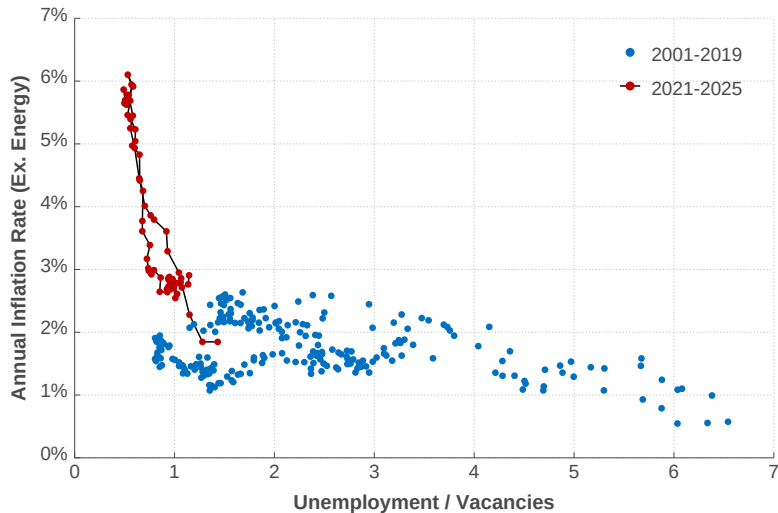
THE LABOR MARKET REMAINS STABLE

THE INFLATION-UNEMPLOYMENT TRADE-OFF (PHILLIPS CURVE): THE CURVE HAS BECOME A DOT



INFLATION VS. LABOR MARKET TIGHTNESS

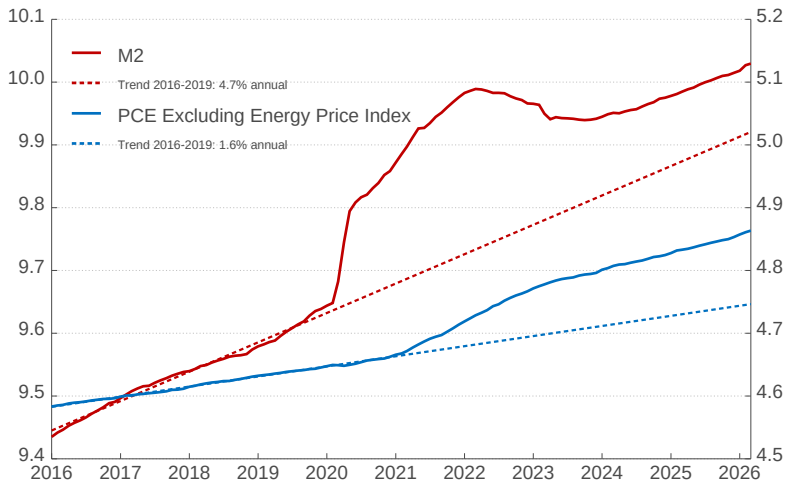
UNEMPLOYMENT-TO-VACANCY RATIO DROPPED BELOW UNITY



$$\mathbf{M} \times \mathbf{V} = \mathbf{P} \times \mathbf{Y}$$

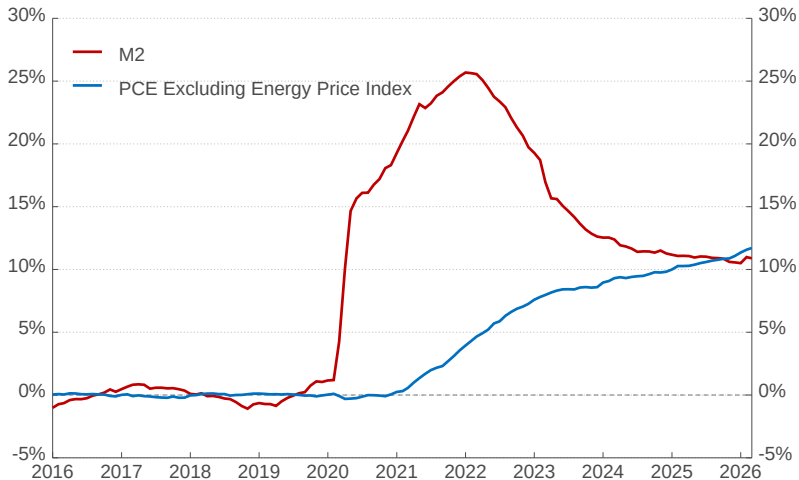
MONEY VS PRICES: MONETARY EXPANSION DURING COVID

IN LOGS: M2 ON LEFT AXIS, PRICES ON RIGHT AXIS—M2 GREW AT SAME RATE AS DISPOSABLE INCOME PRE-COVID



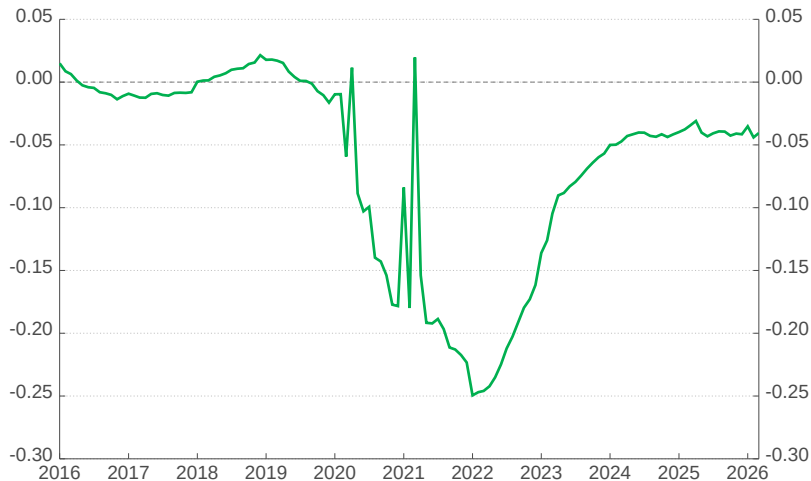
PRICE INCREASE ONLY SLIGHTLY BELOW MONETARY EXPANSION

DEVIATIONS FROM PRE-COVID TREND



STABLE VELOCITY IN 2024-25—SLIGHTLY BELOW PRE-PANDEMIC

$\text{LN VELOCITY} = \text{LN M2} - \text{LN DISPOSABLE PERSONAL INCOME}$ (BASE: 2016-19 = 0)



CONCLUSIONS

In 2025, food, energy and core goods contributing positively to inflation; core services and housing inflation trending downwards.

Trends in prices:

- ▶ food and durable goods show the impact of tariffs
- ▶ nondurables and core services ex. housing have kept a steady growth pace
- ▶ housing significantly moderating its pace, especially since September

Inflation (excluding energy) remains in above-target regime:

- ▶ 2016-19: 1.5%
- ▶ 2021-22: 5.5%
- ▶ 2023-25: 2.8%

Inflation ex tariffs in 2025 estimated at 2.4-2.5%.

Appendix

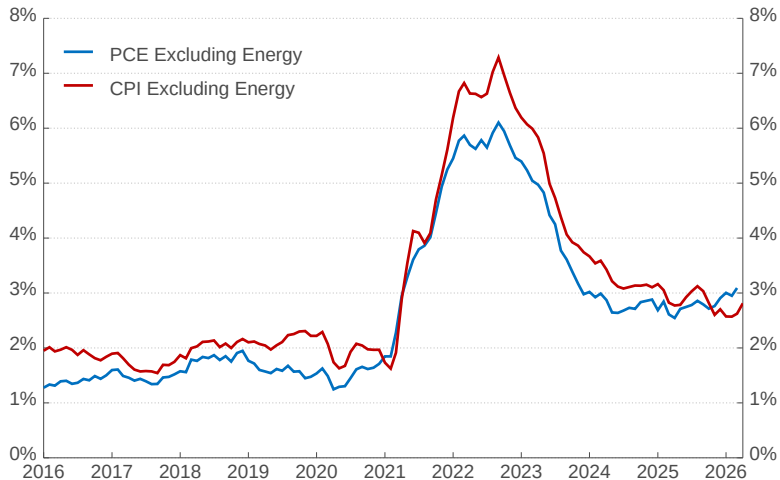
CURRENT INFLATION IN HISTORICAL CONTEXT

PCE EXCLUDING ENERGY ANNUAL INFLATION [▷ BACK](#)



STALLED PROGRESS ON PCE INFLATION WHEN EXCLUDING ENERGY

PCE AND CPI EXCLUDING ENERGY SHOWS LACK OF PROGRESS ON ANNUAL INFLATION [▷ BACK](#)



CPI INFLATION AGGREGATES

PROGRESS PERSISTED LONGER FOR CPI BUT IT HAS ALSO STALLED [▷ BACK](#)

Annualized CPI Inflation Rates

	All Goods & Services	Excluding Energy	Excluding Food & Energy (Core)
2016–19	2.1%	2.0%	2.1%
2020	1.3%	2.0%	1.6%
2021	7.2%	5.6%	5.5%
2022	6.4%	6.4%	5.7%
2023	3.3%	3.7%	3.9%
2024	2.9%	3.1%	3.2%
2025	2.7%	2.7%	2.6%
2026 YTD	6.0%	3.3%	3.3%

SOURCES: FRED, Federal Reserve Board, Federal Reserve Bank of Philadelphia, Bloomberg and author's calculations.

NOTE: The 2026 period is through April; the swaps column remains a manual input.

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NOTE: The

CPI INFLATION COMPONENTS

REVERSALS IN FOOD, ENERGY AND CORE GOODS; MODERATION IN CORE SERVICES AND HOUSING [▷ BACK](#)

Annualized CPI Inflation Rates by Categories

	Food	Energy	Core Commodities	Services Less Rent	Rent of Shelter
2016–19	1.2%	3.9%	−0.2%	2.2%	3.3%
2020	3.9%	−7.0%	1.6%	1.4%	1.9%
2021	6.0%	30.3%	10.7%	3.9%	4.2%
2022	10.1%	6.9%	2.1%	7.5%	7.6%
2023	2.7%	−1.9%	0.1%	3.5%	6.2%
2024	2.4%	−0.3%	−0.7%	4.0%	4.6%
2025	3.0%	2.1%	1.4%	3.4%	3.1%
2026 YTD	3.2%	48.7%	0.8%	4.1%	4.0%

SOURCES: U.S. Bureau of Labor Statistics and author's calculations.

NOTE: The 2026 period is the year through April.

HIGH CORE INFLATION IS WIDESPREAD

ESTIMATED DISTRIBUTION OF CORE PCE INFLATION [▷ BACK](#)

